

LA HW

Included exercises: 1–6, 10–12, 21–23, 26, 29, 32–34

Exercises 1–20: Matrix Transformations

Use the following matrices:

$$A = \begin{bmatrix} 2 & -3 & 1 \\ 4 & 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 5 & 0 \\ 2 & -1 & 3 \\ 0 & 4 & -2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ 0 & -2 \\ 4 & 1 \end{bmatrix}.$$

Original Exercise 1 Give the domain and codomain of the matrix transformation induced by A .

Original Exercise 2 Give the domain and codomain of the matrix transformation induced by B .

Original Exercise 3 Give the domain and codomain of the matrix transformation induced by C .

Original Exercise 4 Give the domain and codomain of the matrix transformation induced by A^T .

Original Exercise 5 Give the domain and codomain of the matrix transformation induced by B^T .

Original Exercise 6 Give the domain and codomain of the matrix transformation induced by C^T .

Original Exercise 10 Compute $T_A \left(\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} \right)$.

Original Exercise 11 Compute $T_B \left(\begin{bmatrix} -4 \\ 2 \\ 1 \end{bmatrix} \right)$.

Original Exercise 12 Compute $T_C \left(\begin{bmatrix} -1 \\ 4 \end{bmatrix} \right)$.

Exercises 21–24: Domain and Codomain Dimensions

Identify the values of n and m for each linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Original Exercise 21

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 \\ x_1 - x_2 \end{bmatrix}.$$

Original Exercise 22

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \\ x_2 \end{bmatrix}.$$

Original Exercise 23

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - 4x_2 \\ 2x_1 - 3x_2 \\ 0 \\ x_2 \end{bmatrix}.$$

Exercises 25–34: Standard Matrices

Compute the standard matrix of each linear transformation.

Original Exercise 26

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 + 3x_2 \\ 4x_1 + 5x_2 \end{bmatrix}.$$

Original Exercise 29

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^4, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ 2x_1 - 3x_2 \\ 0 \\ x_2 \end{bmatrix}.$$

Original Exercise 32

$$T : \mathbb{R}^4 \rightarrow \mathbb{R}^3, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 - x_2 + 3x_4 \\ -x_1 + 2x_4 \\ 3x_2 - x_3 \end{bmatrix}.$$

Original Exercise 33 $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by $T(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^3$.

Original Exercise 34 $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^3$.